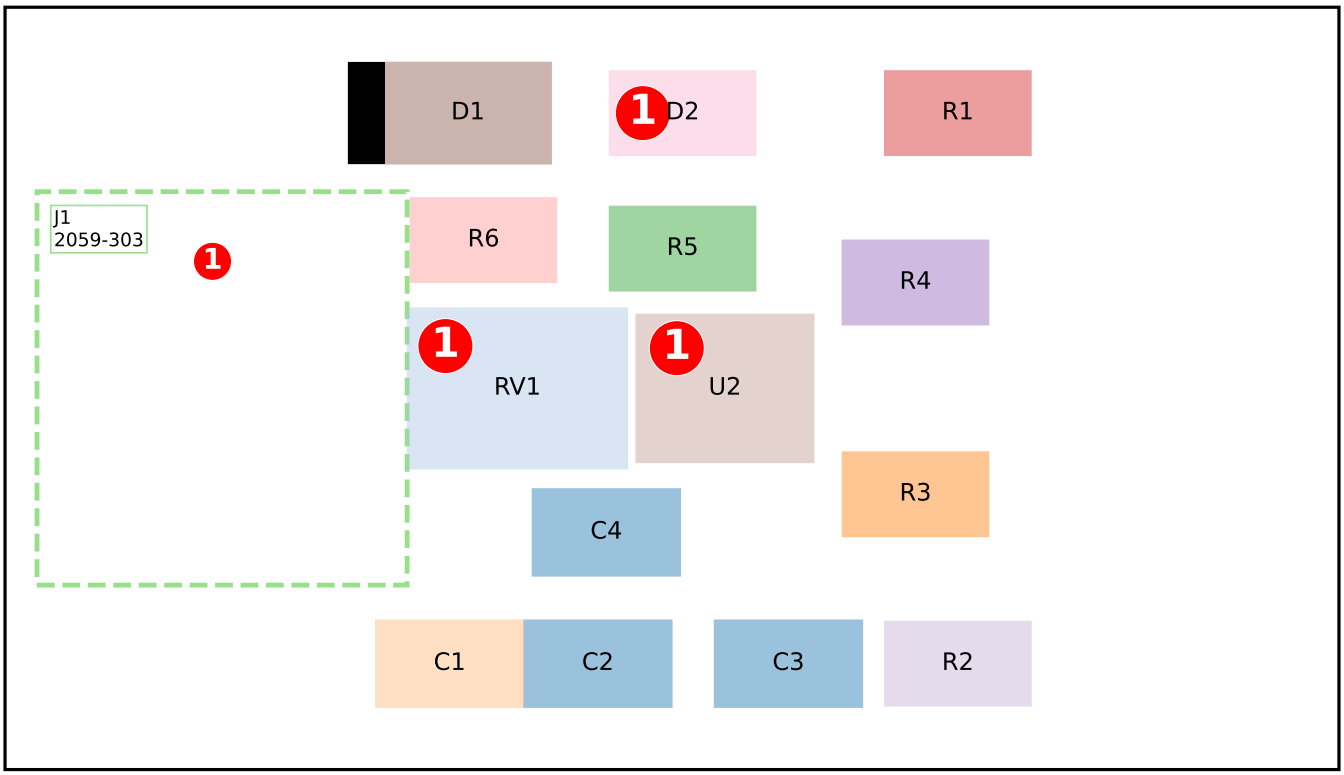
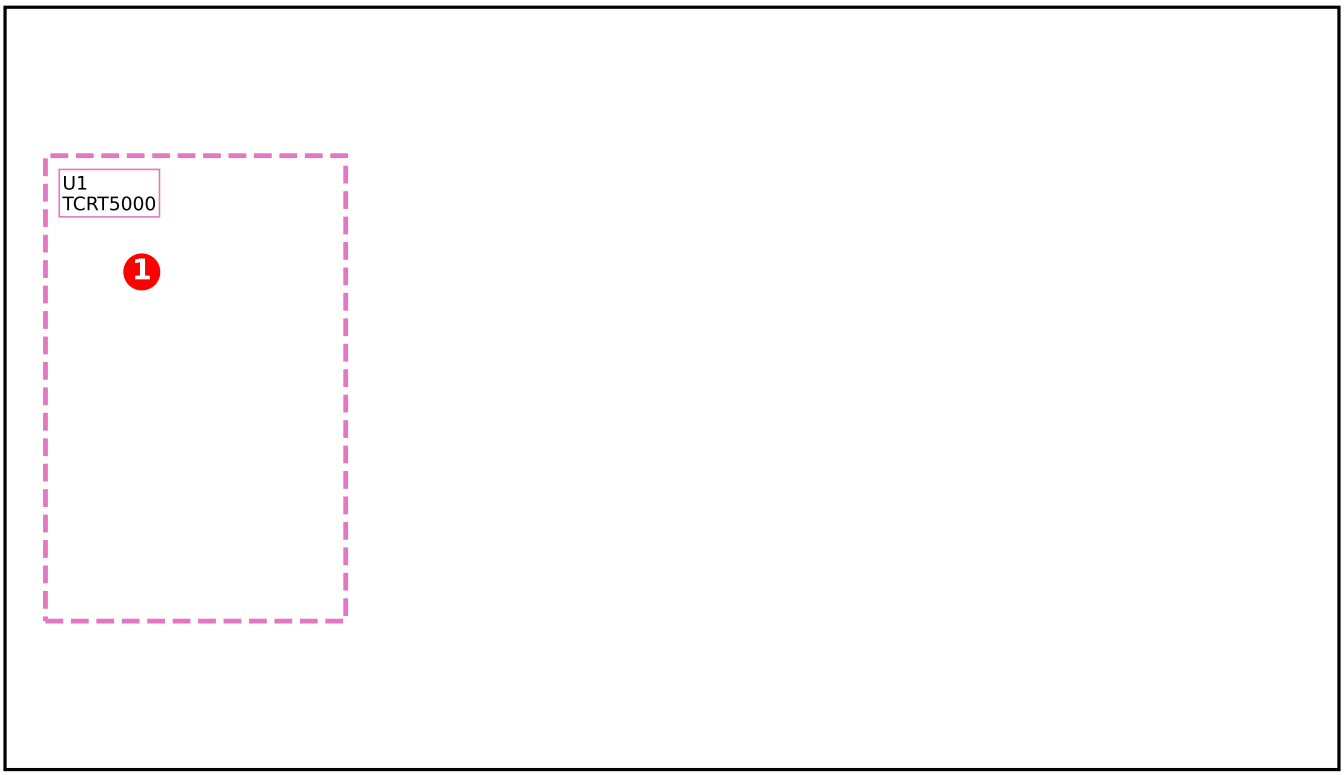
reader board: TOP side

black bar = diode cathode; red + = capacitor +; red 1 = pin 1; black corner = hand-fitted; same colour = same value (values page)



reader board: BOTTOM side (seen from below, mirrored)

black bar = diode cathode; red + = capacitor +; red 1 = pin 1; black corner = hand-fitted; same colour = same value (values page)



reader board: parts by value

100n 50V (0805)	x3	top	-	C2, C3, C4 CC0805KRX7R9BB104 C49678
10k (Potentiometer)	x1	top	pin 1	RV1 TC33X-2-103E C719176
10k (0805)	x1	top	-	R3 0805W8F1002T5E C17414
10u 25V (0805)	x1	top	-	C1 CL21A106KAYNNNE C15850
1k (0805)	x1	top	-	R5 0805W8F1001T5E C17513
2059-303 (TerminalBlock)	x1	top	pin 1	J1 2059-303/998-403 C3309252
220R (0805)	x1	top	-	R1 0805W8F2200T5E C17557
330R (0805)	x1	top	-	R6 0805W8F3300T5E C17630
390k (0805)	x1	top	-	R4 0805W8F3903T5E C17656
4k7 (0805)	x1	top	-	R2 0805W8F4701T5E C17673
B5819W (SOD-123)	x1	top	cathode bar	D1 B5819W C64885
LMV331 (SOT-23-5)	x1	top	pin 1	U2 LMV331IDBVR C34731
TCRT5000 (Vishay)	x1	bottom	pin 1	U1 TCRT5000 C2984661
red (0805)	x1	top	pin 1	D2 NCD0805R1 C84256

reader board: key

Marks on the drawing

Black bar across a diode: the cathode end. Put the band printed on the diode on this side.

Red +: the + pad of an electrolytic or tantalum capacitor.

Red dot with 1: pin 1 of an IC, transistor, LED, connector or module.

No mark: resistors, ceramic capacitors, inductors, fuses, crystals, pin strips and push buttons can go either way round. Bidirectional TVS diodes (part numbers ending in CA) have no polarity either.

Black top-right corner: hand-fitted part (not placed by JLCPCB), e.g. a socketed module or a terminal JLC has no stock of.

The bottom page is mirrored: it shows the board as you see it after flipping it over.

Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pins 1 and 2 are on the opposite side.

SOT-23-5 / TSOT-23-6 / SC-70 / SO / QFN: a dot or a bevelled edge on the body marks pin 1.

Addressable RGB LEDs: one corner of the body or lens is marked (a cut corner or a dot). Which pin it marks differs between makers: check the part's datasheet before soldering, it is easy to misread.

0805 LEDs: pin 1 is the cathode, usually a green line or a T on the underside.

Diodes: the band is the cathode.

Electrolytic capacitors: the stripe on the can marks the - side; the + pad is the other one.

Modules and connectors: match the pin-1 dot to the module's pin-1 marking or to the drawing in its datasheet; for modules on sockets, check the outline (e.g. USB connector towards the board edge).

Tools

0805 parts and SOT-23: a 3-5x magnifier lamp or a 10x loupe to read markings and check joints.

SC-70, QFN and fine-pitch connectors: a USB microscope helps to see bridges and pin-1 marks.

Flux, fine solder (0.5 mm) and a chisel or fine conical tip; tweezers for placing.